

1. DEEP LEARNING – CORE Q&A (BASIC → ADVANCED)

PERCEPTRON

Q1. What is a Perceptron?

Answer:

A Perceptron is the simplest neural network model used for binary classification. It computes a weighted sum of inputs and passes it through an activation function.

Q2. Why is Perceptron limited?

Answer:

- Can only solve **linearly separable problems**
 - Cannot solve XOR problem
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Follow-ups:

- Why does XOR fail?
 - How do we overcome this limitation?
-

ANN (Artificial Neural Network)

Q3. What is an ANN?

Answer:

ANN is a network of interconnected neurons organized in layers (input, hidden, output) that learns patterns from data.

Q4. What is Backpropagation?

Answer:

Algorithm used to update weights by propagating error backward through the network.

Q5. Role of Activation Functions?

Answer:

- Introduce non-linearity
 - Examples: ReLU, Sigmoid, Tanh
-

Follow-ups:

- Why ReLU is preferred over sigmoid?
 - What is dying ReLU?
-

RNN (Recurrent Neural Network)

Q6. What is an RNN?

Answer:

RNN is a neural network designed for sequential data where outputs depend on previous inputs using hidden states (memory).

Q7. Where do we use RNN?

Answer:

- NLP (text, translation)
 - Time series
 - Speech recognition
-

Q8. What is the key difference vs ANN?

Answer:

- ANN → Independent inputs
 - RNN → Sequential dependency (memory)
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Q9. What is Vanishing Gradient Problem?

Answer:

- Gradients shrink during backpropagation
 - Model fails to learn long-term dependencies
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Follow-ups:

- Why does it happen mathematically?
 - How does it affect training?
-

LSTM (Long Short-Term Memory)

Q10. What is LSTM?

Answer:

LSTM is an improved RNN designed to capture long-term dependencies using memory cells and gates.

Q11. Explain LSTM gates

Answer:

- Input Gate → What to store
 - Forget Gate → What to remove
 - Output Gate → What to output
-

Q12. Why LSTM over RNN?

Answer:

- Solves vanishing gradient
 - Captures long-term dependencies
-

Follow-ups:

- What is cell state?
 - Why is LSTM computationally expensive?
-

GRU (Gated Recurrent Unit)

Q13. What is GRU?

Answer:

GRU is a simplified version of LSTM with fewer gates and parameters.

Q14. GRU vs LSTM

Answer:

- GRU → Faster, fewer parameters
 - LSTM → More expressive, better for complex tasks
-

Follow-ups:

- When would you choose GRU over LSTM?
 - Which performs better in real-world?
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COMPARISON (VERY IMPORTANT QUESTION)

Q15. Compare RNN vs LSTM vs GRU

Answer:

Model	Key Feature	Problem Solved	Complexity
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RNN	Sequential memory	Basic sequence	Low
LSTM	Gates + memory cell	Long-term dependency	High
GRU	Simplified gates	Faster training	Medium

Follow-ups:

- Why are Transformers replacing them?
 - When would you still use RNN?
-
-

2. MOCK INTERVIEW SCRIPT (DEEP LEARNING)

ROUND 1: FUNDAMENTALS

Q1. Explain how a neural network learns

Expected Answer:

- Forward pass
 - Loss computation
 - Backpropagation
 - Weight update
-

Follow-ups:

- What if loss is not decreasing?
 - What are common debugging steps?
-

ROUND 2: ARCHITECTURE THINKING

Q2. When would you use:

- ANN
- RNN
- LSTM

Expected Answer:

- ANN → Tabular
 - RNN/LSTM → Sequential
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Follow-ups:

- Why not use ANN for text?

- Why not always use LSTM?
-

ROUND 3: PROBLEM SOLVING

Q3. You are building a chatbot. What model will you use?

Expected Answer:

- LSTM / GRU / Transformer
-

Follow-ups:

- How do you handle long context?
 - What are limitations of RNN-based chatbots?
-

ROUND 4: DEBUGGING

Q4. Model is not learning sequence properly

Expected Answer:

- Vanishing gradient
 - Poor feature representation
 - Sequence length issue
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Follow-ups:

- How to fix?
 - LSTM/GRU
 - Gradient clipping
 - Attention
-
-

3. CASE STUDY (AMAZON-STYLE)

CASE 1: SENTIMENT ANALYSIS (TEXT)

Problem:

Classify movie reviews as positive/negative.

Expected Approach:

1. Clarify

- Length of text?
 - Real-time or batch?
-

2. Data

- Text reviews
 - Labels
-

3. Approach

- Tokenization
 - Embeddings
 - Model → LSTM / GRU
-

4. Metrics

- Accuracy
 - F1 Score
-

5. Challenges

- Long sentences
 - Context loss
-

Follow-ups:

- Why not use simple ANN?
 - How do you handle unseen words?
-
-

CASE 2: TIME SERIES FORECASTING

Problem:

Predict stock prices.

Approach:**Model:**

- RNN / LSTM

Features:

- Past prices
 - External factors
-

Challenges:

- Noise
 - Long-term dependencies
-

Follow-ups:

- Why LSTM better than RNN?
- How do you avoid overfitting?

CASE 3: SPEECH RECOGNITION

Problem:

Convert speech → text

Approach:

- Sequential data
 - Use LSTM/GRU
-

Challenges:

- Variable sequence length
 - Noise
-

Follow-ups:

- Why not CNN?
 - How do you handle accents?
-
-

CASE 4: TEXT GENERATION (ADVANCED)

Problem:

Generate text like ChatGPT (simplified)

Approach:

- RNN → LSTM → Transformer evolution
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Challenges:

- Long context
 - Coherence
-

Follow-ups:

- Why Transformers better?
 - What is attention?
-
-

CASE 5: ANOMALY DETECTION IN LOGS

Problem:

Detect unusual system logs

Approach:

- Sequence modeling
 - LSTM / Autoencoder
-

Follow-ups:

- How do you define anomaly?
 - How do you evaluate?
-

GOLDEN ANSWER FRAMEWORK (DEEP LEARNING)

Candidate should ALWAYS follow:

1. Problem understanding
2. Data type (Tabular / Image / Sequence)
3. Model selection (ANN / CNN / RNN / LSTM / GRU)
4. Challenges
5. Evaluation
6. Production considerations

WHAT INTERVIEWERS CHECK

- Concept clarity (RNN vs LSTM vs GRU)
- Real-world application
- Trade-offs understanding
- Debugging ability